

Continuous Random Variates

Exponential distribution:

$$f(x) = \begin{cases} 1 - e^{-x/\beta} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$f(x) = 1 - e^{-x/\beta}$$

$$\text{or } u = 1 - e^{-x/\beta}$$

$$\text{or } e^{-x/\beta} = 1 - u$$

$$\text{or } -x/\beta = \ln(1-u) \quad [\text{Inverse Transform}]$$

$$\text{or } x = -\beta \ln(1-u)$$

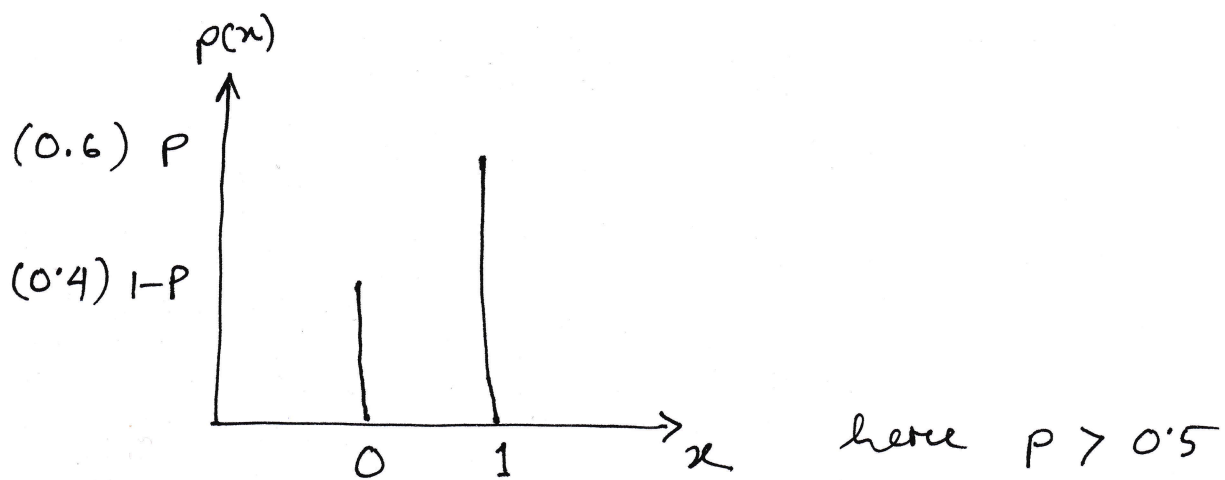
$$X = -\beta \ln U$$

Algorithm:

1. Generate $U \sim U(0,1)$
2. Return $X = -\beta \ln U$

Discrete Random Variates

Bernoulli distribution:



$$p(x) = \begin{cases} P & \text{if } x=1 \\ 1-P & \text{if } x=0 \\ 0 & \text{otherwise} \end{cases}$$

parameter: $p \in (0,1)$

range: $\{0, 1\}$

Expected value: P

Variance: $p(1-p)$

Algorithm:

1. Generate $U \sim U(0,1)$
2. If $U \leq p$, return $X=1$
otherwise, return $X=0$

Expected value:

$$E(X) = \sum x \cdot P(X=x)$$

$$E(X) = (1 \cdot P) + (0 \cdot (1-P))$$

$$\therefore E(X) = P$$

Variance:

$$E(X^2) - \mu^2$$

$$= (1^2 \cdot P) + (0^2 \cdot (1-P)) - P^2$$

$$= 1 \cdot P + 0 - P^2$$

$$= P - P^2$$

$$= P(1-P)$$